Psychophysics

(PCM day 1 session 4)

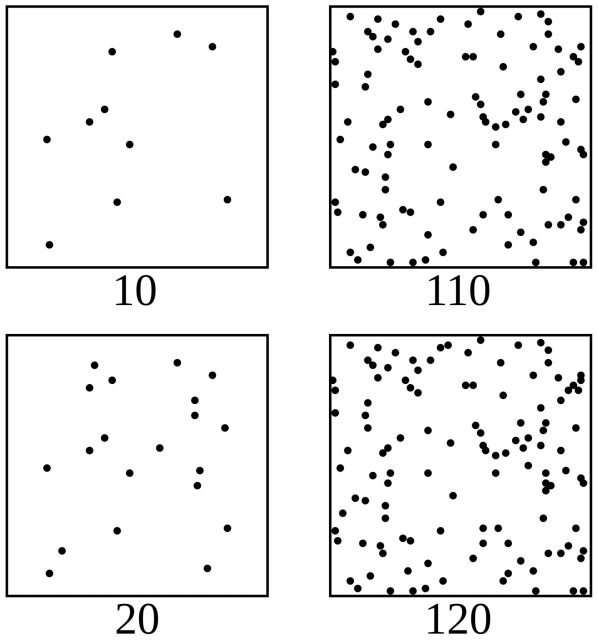
Weber and the ‘just noticeable difference’ (JND)

- Academic at University of Leipzig his entire career
  - became Prof at age 23; retired at age 76
  - beginning in 1834, researched sense of touch

“in observing the disparity between things that are compared, we perceive not the difference between the things, but the ratio of this difference to the magnitude of things compared.”

$$\frac{\Delta R}{R} = k$$

n.b. the constant *k* will be different for different sense modalities



Ernst Heinrich Weber
b. 1795, d. 1878

Image credits: Wikipedia

Fechner and “the Weber-Fechner Law”

- Pupil of Weber, and a better publicist
  - Became Prof. in 1834
  - Five years later, researching afterimages by looking at the sun through colored glasses, damaged his eyes to where he had to resign.
  - After recovery, generalized research on perception to include both luminance and weight

$$p = \alpha \ln \frac{S}{S_0}$$

Gustav Theodor Fechner
b. 1801, d. 1887

But importantly, Weber’s Law and the Weber-Fechner Law were observed to perform poorly when the magnitude of the stimulus was small.

Image credit: Wikipedia

The “Stevens Power Law”

- Politely shreds Fechner starting with paper title, then in article:

To Honor Fechner and Repeal His Law

A power function, not a log function, describes the operating characteristic of a sensory system.

S. S. Stevens

…

Perhaps the most insistent question on this 100th anniversary of Fechner's monumental opus is how its author could have known so much and have made such a wrong guess. (He believed that, unlike errors in general, errors in perception are independent of the perceived magnitude.) Talent, erudition, originality—each of these gifts was his in generous measure, and he applied his skills with signal success to several different domains. Not only did he create psychophysics and pioneer in experimental esthetics, but he also laid the

…

Stanley Smith Stevens
b. 1906, d. 1973

Image credit: Wikipedia

Stevens Law more generally applicable

$$\psi(I) = kI^a$$

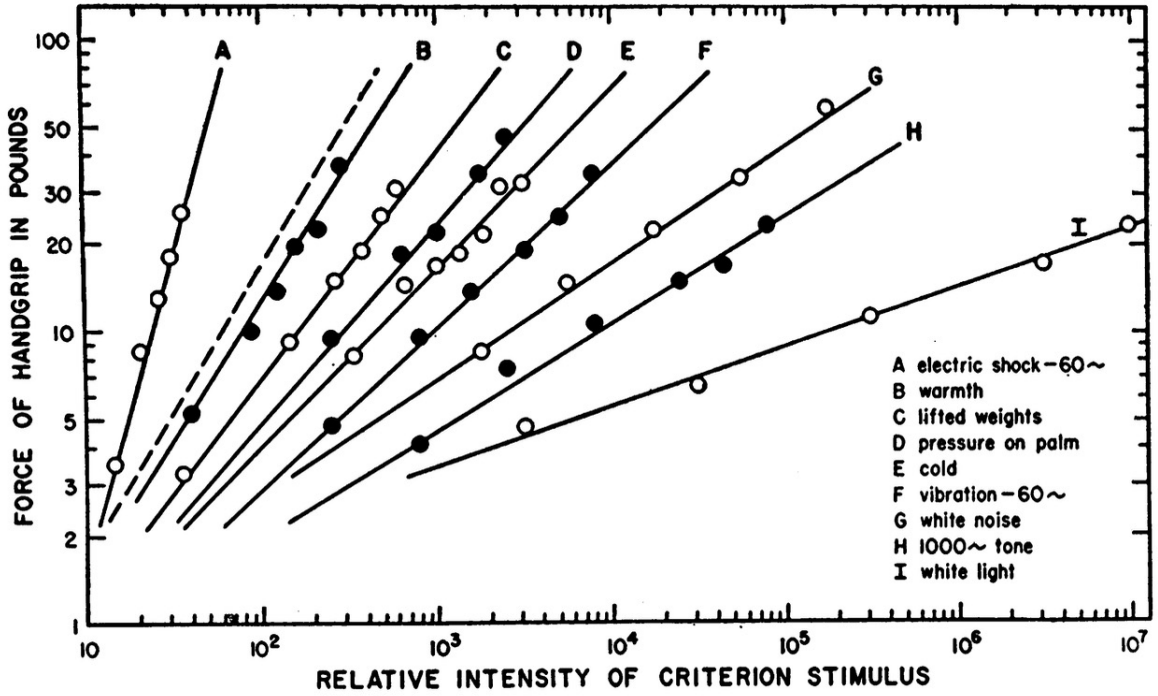


Table 1. Representative exponents of the power functions relating psychological magnitude to stimulus magnitude on prothetic continua.

Continuum	Exponent	Conditions
Loudness	0.6	Binaural
Loudness	0.54	Monaural
Brightness	0.33	5° Target, dark-adapted eye
Brightness	0.5	Point source, dark-adapted eye
Lightness	1.2	Reflectance of gray papers
Smell	0.55	Coffee odor
Smell	0.6	Heptane
Taste	0.8	Saccharine
Taste	1.3	Sucrose
Taste	1.3	Salt
Temperature	1.0	Cold, on arm
Temperature	1.6	Warm, on arm
Vibration	0.95	60 cy/sec, on finger
Vibration	0.6	250 cy/sec, on finger
Duration	1.1	White-noise stimulus
Repetition rate	1.0	Light, sound, touch, shocks
Finger span	1.3	Thickness of wood blocks
Pressure on palm	1.1	Static force on skin
Heaviness	1.45	Lifted weights
Force of handgrip	1.7	Precision hand dynamometer
Autophonic response	1.1	Sound pressure of vocalization
Electric shock	3.5	60 cy/sec through fingers

Image credits: Wikipedia, and original 1961 *Science* article

Weber-Fechner and Stevens for video (1)

What is the k in Weber's law, for video-range luminances?

Keep in mind that this is for SDR video, not HDR video

$$\frac{\Delta R}{R} = k$$

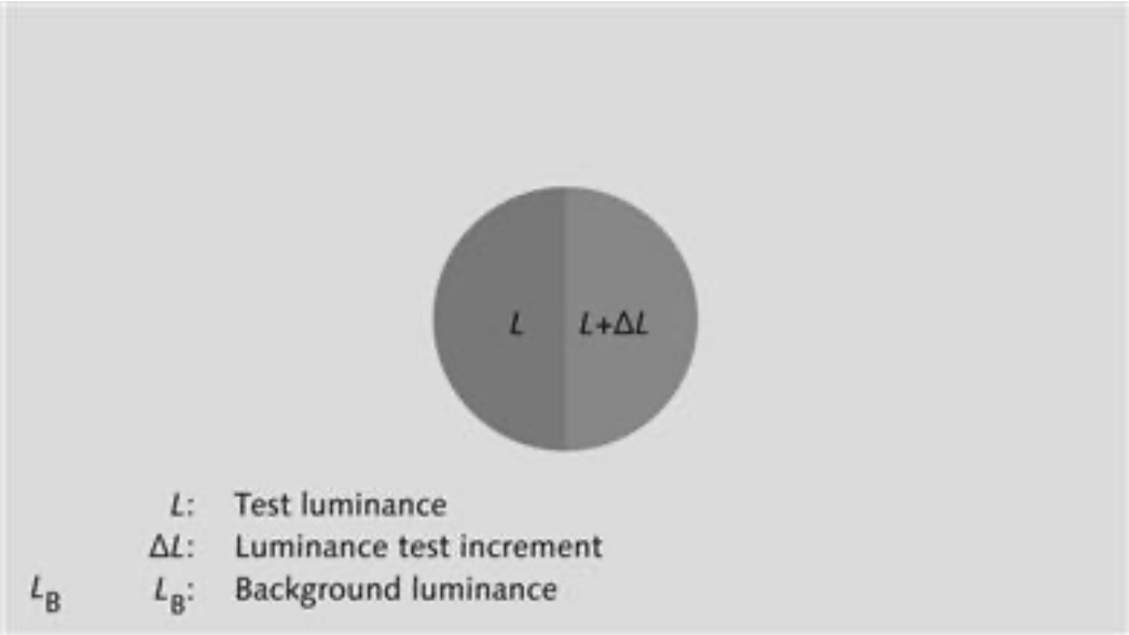


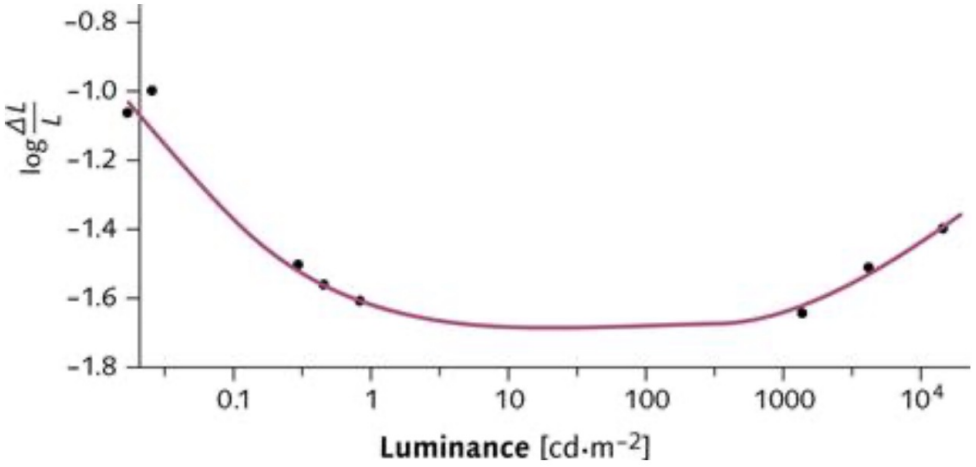
Image credits: Poynton's Digital Video and HD, 2e

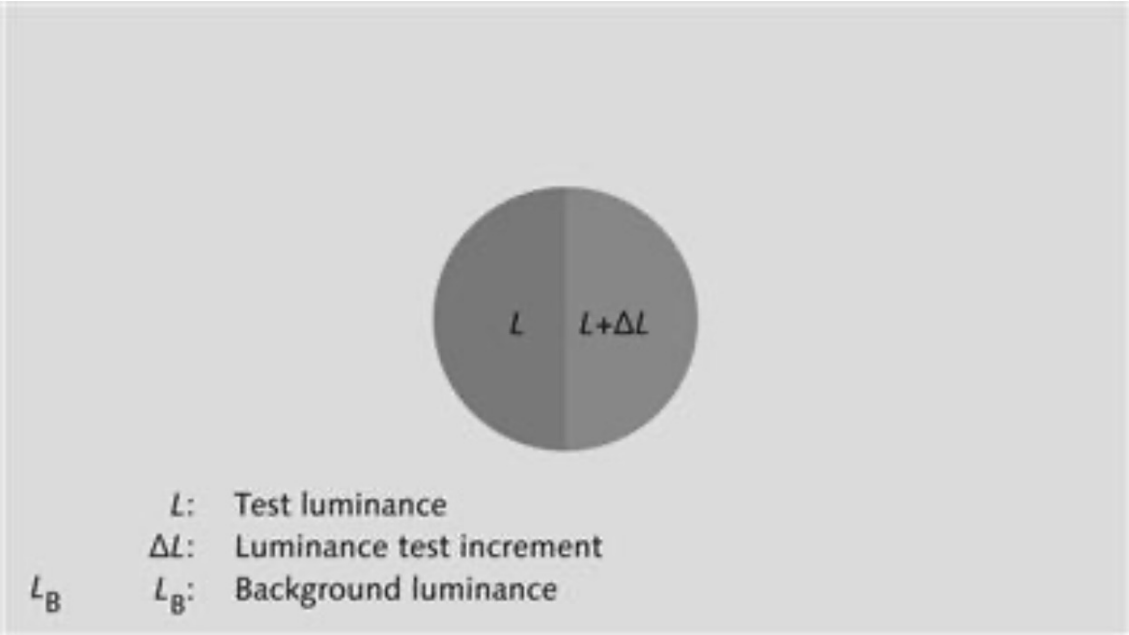
Weber-Fechner and Stevens for video (2)

What is the k in Weber's law, for video-range luminances?

$$\frac{\Delta R}{R} = k$$

Keep in mind that this is for SDR video, not HDR video





For SDR video ranges, *k* is about 0.01, i.e. changing luminance > 1% could produce banding on a ramp (or image)

Image credits: as before, plus Schreiber's 1993 Fundamentals of Electronic Imaging Systems, 3e

“code value 100” problem with dL / L @ 8-bit

- CV 255, 0.3922%
- CV 254, 0.3937%
- …
- CV 101, 0.99%
- CV 100, 1.0%
- CV 99, 1.0101%
- CV 98, 1.0102%

  …
- CV 2, 100%
- CV 1, infinity %
- CV 0

“code value 100” problem with dL / L @ 12-bit

- CV 4095, 0.0244%
- CV 4094, 0.0244%
- …
- CV 101, 0.99%
- CV 100, 1.0%
- CV 99, 1.0101%
- CV 98, 1.0102%

  …
- CV 2, 100%
- CV 1, infinity %
- CV 0

“code value 100” problem with dL / L @ 16-bit

- CV 65535, 0.002%
- CV 65534, 0.002%
- …
- CV 101, 0.99%
- CV 100, 1.0%
- CV 99, 1.0101%
- CV 98, 1.0102%

  …
- CV 2, 100%
- CV 1, infinity %
- CV 0

”code value 100” problem dynamic range impact

- @ 8bit: can use 100 to 255, so log2(255/100) = 1.35 stops
- @ 12bit: can use 100 to 4095, so log2(4095/100) = 5.36 stops
- @16bt: can use 100 to 65535, so log2(65535/100) = 9.36 stops

- point of comparison: ALEXA 35 records about 17 stops

real-world luminance encodings

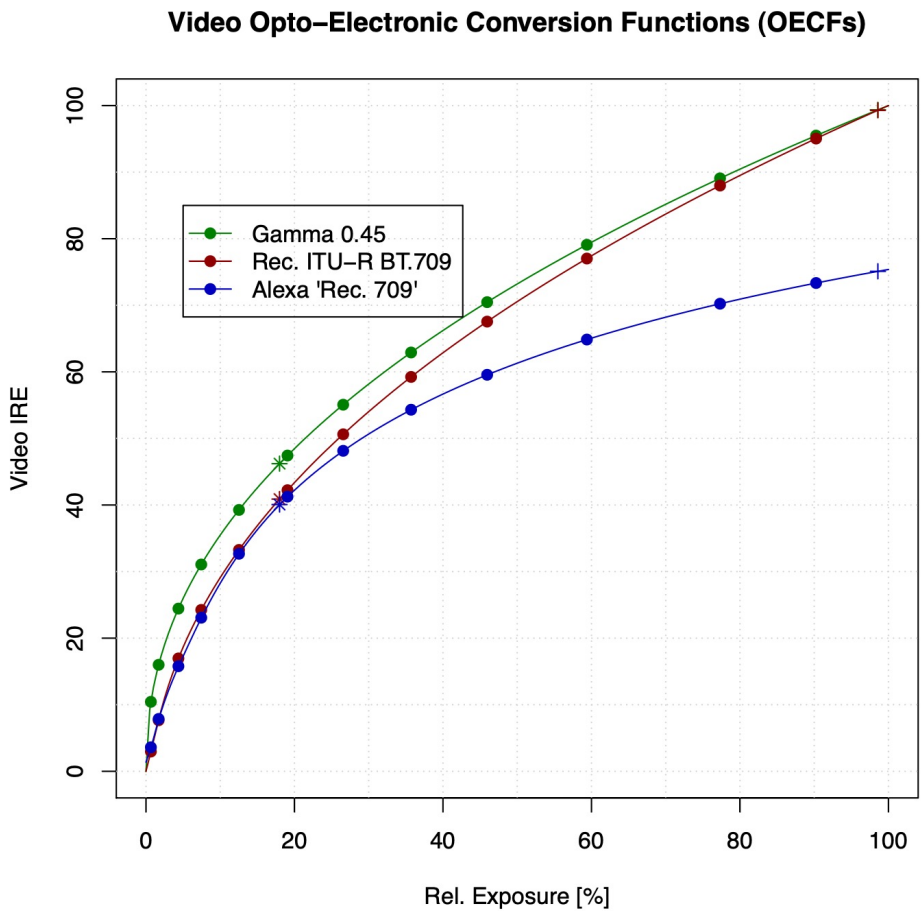
- above the darks, Weber-Fechner applies, and is logarithmic
- down in the darks, Stevens Law applies and is a power function
- most real-world luminance encodings are hybrids, sometimes with linear portions for the darkest luminances.

1 Opto-electronic conversion

Item	Parameter	System Values
1.1	Opto-electronic transfer characteristics before non-linear pre-correction	Assumed linear
1.2	Overall opto-electronic transfer characteristics at source(1)	$V = 1.099\ L^{0.45} - 0.099$ for $1 \geq L \geq 0.018$ $V = 4.500\ L$ for $0.018 > L \geq 0$ where: L : luminance of the image $0 \leq L \leq 1$ V : corresponding electrical *signal*

Pure gamma vs. Rec. 709 vs. ALEXA direct 709

- A ‘pure’ 0.45 gamma would have an extremely steep start, so much so that it would waste code values at the bottom of the encoding range
- Thus the linear segment on the last page.
- Note that both the pure 0.45 gamma encoding and the BT.709 encoding basically use up all their code values by the time they get to a 100% white reflector; whereas the ALEXA output can handle much, much brighter objects.



Last bit: why is 18% called ‘middle gray’?

- 18% reflectance turns into 50% of the encoded signal
  - or close enough: try raising 0.18 to 0.454545… and see where you end up.

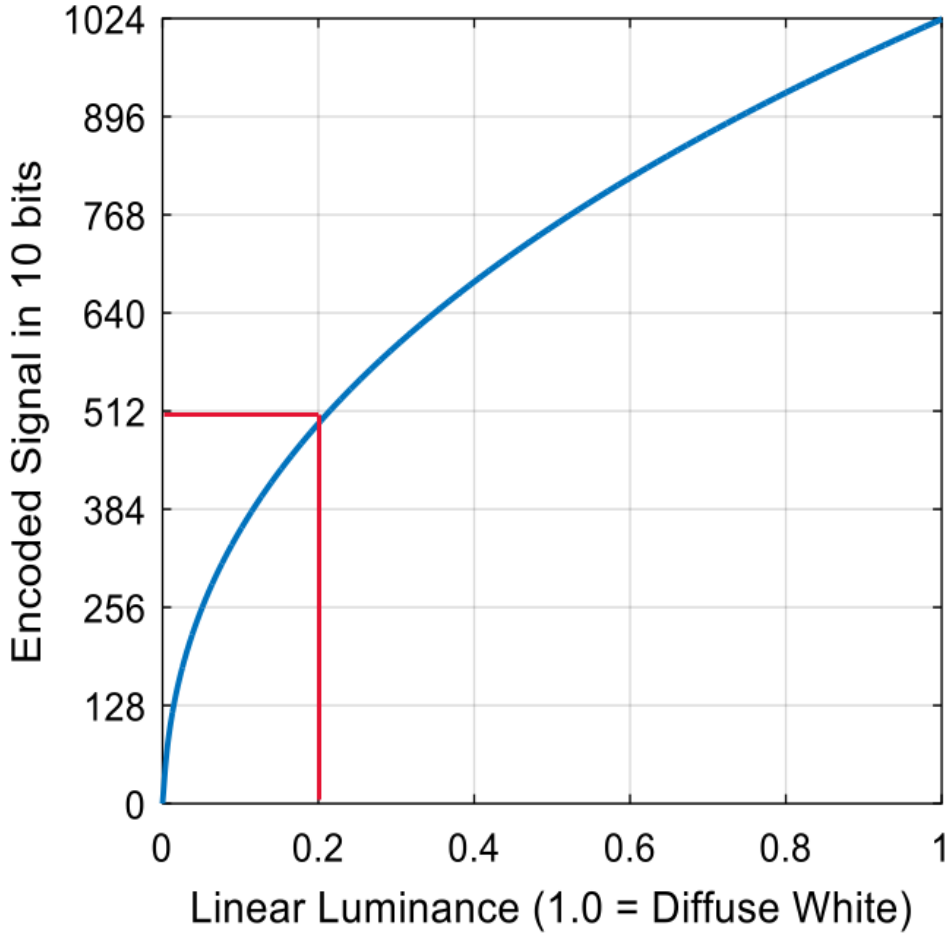


image source: Prof. Jan Frölich

next up (tomorrow):

- color reproduction technologies

- questions?